세브란스병원
SEVERANCE HOSPITAL

항암약물요법 투여 시 주의사항 및 부작용 발생 시 대처방안

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	일반명	상품명	함량 규격	계열	Extravasation	투여시 주의 사항	모니터링	부작용 및 발생시 대처방안	Decontamination	Elimination
1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		
2	Amivantamab	**RYBREVANT**	350mg/7ml/V	EGFR, MET 표적으로 하는 이중 특이성 항체(bispecific antibody)		1. 유량 조절기와 in-line 무균, 무발열성, 저단백질 PES 필터(0.2㎛)를 끼운 주입세트를 이용하여 희석 용액을 정맥 내 투여 2. 투여 세트의 재질: 폴리우레탄(PU, 폴리부타디엔(PBD)	EGFR exon 20 insertion mutation status, pregnancy status, signs/symptoms of dermatologic toxicity, ocular toxicity, signs/symptoms of infusion reactions during infusion, HBV screening, Cardiovascular monitoring	과량투여 발생시 이 약 투여를 중단하고, 임상적인 독성이 감소 또는 해결될 때까지 일반적인 보조요법을 시행		
3	Aflibercept	**ZALTRAP**	100mg/4ml/V 200mg/8ml/V	rhMab(anti VEGF)		1. 1hr(IVF) 2. IV push불가, 유리체내 주사 불가(고삼투압)	CBC , Urine protein, Blood pressure, Diarrhea and dehydration, Pregnancy test, Signs & symptoms of	과량투여시 적절한 처치, 특히 고혈압과 단백뇨를 모니터링하고 치료해야 하며, 이상반응을 모니터링하기 위한 면밀한 의학적 관리가 필요		
4	Arsenic Trioxide	**TRISENOX**	12mg/6ml/V	Arsenic Trioxide	irritant, 냉찜질	2시간에 걸쳐 IVF	Serum electrolytes, Hematologic and coagulation profiles, Glucose levels, Hepatic profiles, Creatinine, ECG	과량 투여시 즉시 chelation therapy : BAL, succimer	즉시 chelation therapy - INHALATION EXPOSURE: 노출된 약으로부터 환자 격리하고 24시간 urine 검사하여 chelate 여부 판단 - DERMAL EXPOSURE: 다른 부위 접촉을 피해 철저하게 세척, 국소 steroid cream - EYE EXPOSURE: 대량의 물로 irrigation, 안과	혈액투석 고려
5	Atezolizumab	**TECENTRIQ**	840mg/14ml/V 1,200mg/20ml/V	Anti-PD-L1 Monoclonal		60min(IVF) bolus 투여 및 IVH 금지, 첫 주입에 내약성 보이는 경우 30분	Monitor liver function tests (AST, ALT, and bilirubin; at baseline and periodically during treatment), thyroid			
6	Avelumab	**BAVENCIO**	200mg/10ml/V	Anti-PD-L1 Monoclonal Antibody		IVF 1. IV push 불가 2. 투여시 0.2 μm 크기의 멸균, 비발열성, 저단백결합 인라인 필터 사용	Cardiovascular events, Hyperglycemia, Diabetes, Liver enzymes, Serum creatinine, Infusion-related reactions(pyrexia, chills, flushing, hypotension, dyspnea, wheezing, back pain, abdominal pain, and urticarial), Underlying immune-mediated adverse reactions, Thyroid function, Transplant-related complications	과량투여시, 약물이상반응의 징후나 증상에 대해 환자를 면밀히 모니터링하고, 적절한 대증치료를 실시해야 한다.		
7	Azacitidine	**AZALID**	150mg/V(P)	Pyrimidine analogue		1. IVF - 10~40분에 걸쳐 주입 - 조제후 실온 1시간내에 투여 완료 - D5W, 헤스판 및 중탄산염 함유용액 혼합금기(불안정) - IV 투여시 혈액학적 독성 증가하나 투여 가능 2. SC - 4ml이상 용량은 2개의 시린지에 나누어 투여	CBC, Measurable residual disease (MRD), Liver chemistries, Renal function, including a serum creatinine and electrolytes, Serum bicarbonate level, Tumor lysis syndrome, Pregnancy test	과량투여시 적절한 혈구수인지 모니터하고 필요하다면 보존적 치료 필요	피부노출시 비누와 물로 세척, 점막에 노출 시 물로 씻어냄	
		VIDAZA	100mg/V(P)			1. IVF - IV 투여시 혈액학적 독성 증가하나 투여 가능	CBC, serum bicarbonate, hepatic function, renal function tests and signs of renal toxicity	1.Hypotensive episode: IV 0.9% NaCl at 10 mL to 20 mL/kg, dopamine, norepinephrine.		Early hemodialysis 효과적
8	Belotecan	**CAMTOBELL**	2mg/V(P)	S phase, Plant		30min(IVF) 주사시 차광, 투여 권장 안정성 4 hr			피부접촉시 물과 세제로 세척, 점막에 노출 시 물로 씻어냄	
9	Bendamustine	**BENCORD**	25mg/V(P) 100mg/V(P)	Alkylating	Irritant (vesicant-like properties)	30-60min(IVF)	Leukocytes, platelets, Hb, neutrophil counts, blood chemistries, signs and symptoms of tumor lysis	혈액학적 이상반응을 관리하는 효과적인 대처법은 골수 이식 및 수혈(혈소판, 농축적혈구) 또는 혈액성장인자 투여	접촉한 신체부위는 비누 및 물로 조심해서 헹궈내야 하며 눈은 생리식염수로 헹궈냄	
10	Bevacizumab	**AVASTIN** **ONBEVZI**	100mg/4ml/V 400mg/16ml/V	rhMab(anti VEGF)		IV push불가	Proteinuria, blood pressure, infusion reation	Severe hypertension: nitroprusside 투여		
11	Bleomycin	**BLEOMYCIN**	15mg/V(P)	G2,M phase, Antibiotic	Irritant, 냉찜질	IV, IVF, IM, SC 1. Test dose(1-2unit), D5W 희석시 24시간내 10-16%손실, 유리병(PVC bag장기간주입 피할것) 2. 400mg (pulmonary toxicity)	Chest roentgenograms, renal function, pulmonary diffusion capacity for carbon dioxide	Hypotension : IV 0.9% NaCl, dopamine, norepinephrine Bleomycin-induced pulmonary toxicity: Corticosteroid	피부접촉시 물과 비누로 세척, 점막에 노출 시 물로 씻어냄	
12	Blinatumomab	**BLINCYTO**	35μg/V(P)	Bispecific Antibody		1. IVSS주입하여 infusion bag으로 만듬. 2. 1cycle:(D1-7):9μg/day,(D8-28)28μg/day 2cycle:28μg/day 3. 0.2-1.2 μm 크기의 무균,비발열성,저단백결합 인라인 필터사용	CBC, AST,ALT, gamma-glutamyl transferase, and total blood bilirubin, neurotoxicity, DIC, capillary leak syndrome, and hemophagocytic lymphohistiocytosis/macrophage activation syndrome			
13	Bortezomib	**VELCADE** **VELKIN**	3.5mg/V(P) 2.5mg/V(P)	Proteasome Inhibitor	Irritant, 냉찜질	연속 투여 시 적어도 72시간의 간격 필요 1. IV: 3~5초 IV injection	CBC, platelet count, hepatic function, blood glucose levels, signs and symptoms of peripheral neuropathy,	Hypotensive episode: IV 0.9% NaCl, dopamine, norepinephrine		
14	Brentuximab	**ADCETRIS**	50mg/V(P)	CD30-directed ADC	Irritant, 냉찜질	1. 1회 최대용량 180mg 2. IVF(30min이상)	CBC, Liver funtion, Peripheral neuropathy, tumor lysis syndrome, Symptoms of CNS system abnormalities			
15	Busulfan	**BUSULFEX**	60mg/10ml/V	Alkylating, CCNS	Irritant 냉찜질	1. IVF(2hr) 2. Polycarbonate syringes or polycarbonate filter needle 사용금지	CBC, Serum transaminases, alkaline phosphatase, bilirubin, Seizures, Symptoms of cardiac tamponade	Seizure: IV benzodiazepines, barbiturates	피부나 점막에 접촉되면 즉시 물로 세척	Hemodialysis or hemoperfusion 고려
16	Cabazitaxel	**JEVTANA**	60mg/1.5ml/V	Mitotic Inhibitor	Vesicants (Non-DNA-binding), 온찜질	IVF(1hr이상) 1. 0.22 마이크로미터인 직렬필터 사용 2. Premedicate with an antihistamine, a corticosteroid, and an H2 antagonist at least 30 minutes prior to infusion. 3. 조제시 PVC재질 용기 사용금지	CBC with differential, signs of hypersensitivity reaction	1. Nausea/vomiting, diarrhea: IV fluids, antidiarrheals (eg, high-dose loperamide, diphenoxylate) and antiemetics. 2. Hypersensitivity reaction: - MILD/MODERATE: Antihistamines with or without inhaled beta agonists, corticosteroids or epinephrine. - SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, and IV fluids	피부접촉시 비누와 물로 세척, 점막에 노출 시 물로 씻어냄	Hemodialysis, hemoperfusion and plasmapheresis 권장되지 않음
17	Carboplatin	**CARBOPLATIN** **NEOPLATIN**	600mg/60ml/V 150mg/15ml/V 450mg/45ml/V	Alkylating	Irritant, 냉찜질	1. 주사시 차광 2. Al과 반응하여 침전(Al함유의료기기 사용 금지) 3. S포함하는 아미노산 수액과 배합금기	CBC(including differential)	1. Nephrotoxicity: NS 3~6 L/day로 적극적인 IV fluid resuscitation(목표 urine output: 1 to 3 mL/kg/hr) 2. Hypersensitivity reaction:	피부접촉시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	응급 혈액투석 실시

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		
18	Carfilzomib	**KYPROLIS**	30mg/V(P)	2nd gen. Proteasome Inhibitor	Non-vesicants	1. 거품주의(흔들지말것) 2. BSA > 2.2인 경우 2.2 기준 투여. Regimen및 Cycle에 따른 투여용량 유의할것	CBC with differential and platelets (monitor frequently throughout therapy), serum potassium levels regularly during treatment, renal function, pulmonary function (with new or worsening pulmonary symptoms), liver function tests, blood pressure. Signs/symptoms of infusion-related reactions, congestive heart failure, tumor lysis syndrome, peripheral neuropathy, posterior reversible encephalopathy syndrome, thrombocytopenic thrombotic purpura/hemolytic uremic syndrome, and venous thromboembolic events. Monitor for evidence of			
19	Carmustine	**CARMSPAL-P***	100mg/V(P)	Alkylating	Vesicants (DNA binding), 냉찜질	최대 주입 속도(3 mg/m(2)/min)로 투여시, 천천히 적어도 2 hours 이상 IV infusion	Hematocrit, platelet count, total leukocyte count with differential, renal function, liver enzymes, and respiratory status	Localize by applying dry cold compresses for 20 minutes 4 times daily for 1 to 2 days. Avoid alcohol compresses. Elevate the limb. Administer analgesia if needed	피부노출 시 일시적인 hyperpigmentation 유발 가능. 노출 시 즉시 비누와 물로 완전히 세척	
20	Cetuximab	**ERBITUX**	100mg/20ml/V	rhMab (anti EGFR)		1. IV push or bolus 투여 금지 2. 0.22-micrometer inline filter 사용하여 투여	Serum electrolytes, CBC, ECG, hepatic enzyme, signs of severe infusion reactions	1. Hypotensive episode: 0.9% IV NaCl, dopamine, norepinephrine -Mild hypotension: SC epinephrine -Severe hypotension : IV epinephrine 2. Anaphylaxis - Bronchospasm: Oxygen, antihistamines, inhaled beta agonists - Upper airway edema: SC epinephrine		
21	Cisplatin(DDP)	**CISPLAN**	10mg/20ml/V 50mg/100ml/V	Alkylating 유사 (DNA cross-linking	Irritant, 냉찜질	1. 주사시 차광 2. AI와 반응하여 침전(AI함유의료기기 사용 금지)	CBC, hepatic function, renal function, audiometric testing, neurologic exam, symptoms of anaphylactic	Antidote: Sodium thiosulfate 1. Nephrotoxicity:	피부접촉시 비누와 물로 세척, 점막에 노출 시 물로 씻어냄	1-2시간 이내 sodium thiosulfate 투여. large
22	Cladribine	**LEUSTATIN**	10mg/10ml/V	Antimetabolite, S phase	Non-vesicants	1. 0.22 micron filter 사용 2. 조제 후 투여 지연되는 경우 냉장 8hr까지만 보관가능	CBC with differential, serum electrolytes, hepatic enzymes and renal function, signs and symptoms of tumor lysis syndrome, fever and signs of infection, peripheral neuropathy			
23	Clofarabine	**EVOLTRA**	20mg/20ml/V	Antimetabolite	Non-vesicants	1. Cytarabine과 병용 시 evoltra 투여 후 4시간 후에 투약 2. Na 70.77mg (3.08mmol)함유, Na 식이조절 환자 고려 3. 치료 5일 동안 continuous IV fluids infusion 4. 0.22㎛ in-line filter를 연결한 후 희석수액병에 주입	CBC and platelets, hepatic and renal function, cardiac function, signs and symptoms of venoocclusive disease of liver, infection, tumor lysis syndrome, cytokine releas, systemic inflammatory response syndrome and capillary leak syndrome	1. Hypotensive episode: IV 0.9% saline (성인: 1~1.5 L, 소아: 20 mL/kg). 대량의 IV fluids는 오히려 capillary leak syndrome 악화. hypotension 지속시 vasopressors 투여. 2. Systemic inflammatory response syndrome: Hydrocortisone 100 mg/m(2) for 3 days is recommended		
24	Cyclophosphamide	**ENDOXAN**	500mg/V(P)	Alkylating, CCNS	irritant, 냉찜질	IV, IVF 25℃ 이하에서 보관	CBC, cardiotoxicity, urinary sediment	1. Bladder toxicity: hydration, MESNA투여, formalin 5-10%의 방광내 주입, hyperbaric oxygen therapy 2. SIADH: fluid restriction, furosemide	피부접촉시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	투석가능 (20% to 50%)
25	Cytarabine	**CYTARABINE(Faulding)**	1000mg/10ml/V 2000mg/20ml/V	Pyrimidine analogue, S phase	Irritant or neutral, 냉찜질	IV, IVF, SC, IT Heparin 배합불가	CBC, renal and hepatic function, serum uric acid concentrations, neurologic and GI function	1. Diarrhea: - Mild to moderate (Grade 1~2) without infection: loperamide(2 mg every 4 hours for 24 hours) - Progressive, uncontrolled: octreotide, IV fluids, and antibiotics 2. Seizure: IV benzodiazepines, barbiturates 3. Acute lung injury: Maintain adequate ventilation and oxygenation 4. Intrathecal injection:	피부접촉시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Early hemodialysis 효과적
		CYTARABINE	100mg/2ml/V 2000mg/40ml/V	Antimetabolite, S phase						
	Liposomal cytarabine	**DEPOCYTE***	50mg/5ml/V	Anti-SLAMF7	irritant, 냉찜질	IT only(1-5min 천천히) 1. 보존제 미함유로 조제 후 4시간 안에 사용, 2. In line 필터 사용금지. 3. 5일간 dexamethasone 4mg bid 투여	Hematopoietic system, neurotoxicity, especially within 5 days of administration			
26	Dacarbazine	**DTI**	100mg/V(P) 200mg/V(P)	Alkylating, CCNS	Irritant, 냉찜질	1. IVF(30min이상) 2. 주사시 차광	CBC	1. Seizure: IV benzodiazepines, barbiturates 2. Anaphylaxis:		
27	Daratumumab	**DARZALEX**	100mg/5ml/V 400mg/20ml/V	Mab(anti CD38)		투여시 0.2-0.22 μm 크기의 멸균,비발열성,저단백결합 인라인 PES필터 사용	Blood screen and type testing, CBC, Signs and symptoms of infusion reactions			
28	Dactinomycin	**DACTINOMYCIN**	0.5mg/V(P)	Antibiotic	Vesicant, 냉찜질	IV, IVF 6개월 미만 소아 사용 금지	Renal and hepatic function, bone marrow function	냉찜질: Localize by applying drug cold compresses for 20 minutes 4 times daily for 1 to 2 days. Avoid alcohol compresses. Elevate the limb and administer analgesia if needed Seizure: IV benzodiazepines, barbiturates	가루나 증기를 흡입하거나 피부 또는 눈과 같은 점막에 닿았을 때 눈의 경우 즉시 15분 이상 충분한 물로 씻어낸 후 안과진찰, 피부 접촉시 15분 이상 충분한 물로 씻어낸 후 진찰	
29	Daunorubicin	**DAUNOCIN**	20mg/V(P)	Antibiotic	Vesicant, 냉찜질	IV, IVF 주사시 차광, 뇨적색	CBC, hepatic, renal, cardiac function	Cardiomyopathy 예방 및 extravasation 치료: Dexrazoxane Remove cold compress at least 15 minutes prior to dexrazoxane administration	피부접촉시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Early Plasma exchange 고려. 투석 권장하지 않음
30	Decitabine	**DACOGEN**	50mg/V(P)	Pyrimidine analogue		IVF(1hr) 희석 후 15분 이내 사용, 7hr까지 냉장보관가능	CBC, liver function test, serum creatinine, symptoms of infection	과량투여의 경우 표준 보조 수단: 호중구감소증과 혈소판감소증을 포함한 골수억제의 증가와 관련		
	Decitabine	**DECILID**	40mg/V(P)			IVF(1hr)	CBC, Platelet count, Liver function tests, Pregnancy test, Serum creatinine			
31	Dinutuximab beta	**QARZIBA***	20mg/4.5ml/V	Anti-GD2		1. 1% human albumin을 포함한 0.9% NS에 희석 2. 0.22 μm 인라인 필터 사용, direct sunlight, heat 노출금지	CBC, serum electrolytes, oxygen saturation, signs/symptoms of infusion reactions, pain, peripheral neuropathy, capillary leak syndrome, infection/sepsis, hemolytic uremic syndrome, ocular toxicity, urinary retention, transverse myelitis, and/or reversible posterior leukoencephalopathy syndrome			
32	Docetaxel	**DITAXEL1-VIAL**	20mg/1ml/V 80mg/4ml/V 120mg/6ml/V	Antimicrotubule (M phase)	irritants, 냉찜질	IVF(1hr) 1. in-line filter set 사용 불필요 2. PVC 용기사용 불가(DEHP용출)	CBC, hepatic function, ECG, hypersensitivity reactions, preexisting effusion exacerbation	Antidote: Hyaluronidase Inject 150 to 900 international units subQ around the extravasation area 1. Myalgias: Initially treat with NSAIDs 2. Hypersensitivity reaction:	피부접촉시 비누와 물로 세척, 점막에 노출 시 물로 씻어냄	투석 권장하지 않음
		TAXOTERE ONE	20mg/1ml/V 80mg/4ml/V							
33	Doxorubicin	**ADRIAMYCIN PFS**	10mg/5ml/V 50mg/25ml/V	S phase, Antibiotic	Vesicant, 냉찜질	IVF(1hr), IV 1. SC/IM 금기, Heparin 배합금기 2. 뇨적색 3. 550mg/m^2 (IVH, w/o RT)	CBC, hepatic function, blood uric acid, creatinine levels, cardiac function(LVEF)	Cardiomyopathy 예방: dexrazoxane 투여 Remove cold pack at least 15 minutes prior to dexrazoxane administration	피부접촉시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	early hemoperfusion이 배설증가시킴. 투석 권장하지 않음
		AD MYCIN	10mg/5ml/V 50mg/25ml/V							

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		
34	Doxorubicin HCl liposomal	**CAELYX**	20mg/10ml/V	S phase, Antibiotic	Irritant, 냉찜질	IVF(60min이상), 주입 완료까지 차광 1. 연속 투여 시 본제 먼저 투여 후 Paclitaxel 투여 2. In line filter 사용 금지 3. IV/SC/IM 금기, Heparin 배합금기 4. 한계누적량 550mg/m^2	CBC, hepatic function tests, blood uric acid, potassium, calcium, phosphate, and creatinine levels, left ventricular ejection fraction	Apply cold pack for 15 to 20 minutes at least 4 times a day for first 24 hours Cardiomyopathy 예방: dexrazoxane 투여	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Early hemoperfusion 고려. Dialysis 권장되지 않음
35	Durvalumab	**IMFINZI**	120mg/2.4ml/V 500mg/10ml/V	Anti-PD-L1 Monoclonal		IVF(60min) 투여시 0.2-0.22 μm 크기의 멸균,저단백결합 인라인 필터 사용	Chemistry, Liver function, Thyroid function, Blood glucose, Renal function, Signs or symptoms	과량투여 사건의 경우, 의사는 일반적인 보조 조치를 따라야 하며 대증치료를 실시해야 한다.		
36	Elotuzumab	**EMPLICITI***	300mg/V(P) 400mg/V(P)	Anti-SLAMF7		1. 0.2-0.22 μm 크기의 멸균,저단백결합 인라인 필터 사용 2. 재구성 시 세게 흔들지 말것. 용해 후 5분 방치	Liver enzymes, periodically, Signs and symptoms of infection, Development of second primary malignancies			
37	Eribulin mesylate	**HALAVEN**	1mg/2ml/V	Mitotic Inhibitor	Non-vesicants	IV, IVF 1. D5W에 희석 불가 2. syringe에서 실온 4hr, 냉장 24hr 안정	CBC, serum electrolytes, renal function, liver enzymes, ECG, peripheral motor or sensory neuropathy	Hypersensitivity reaction: -MILD/MODERATE: Antihistamines with or without inhaled beta agonists, corticosteroids or epinephrine -SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, IV fluids	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Dialysis, hemoperfusion, plasmapheresis 권장되지 않음
38	Etoposide(VP-16)	**E.P.S**	100mg/5ml/V	G2 phase, Plant	Irritant, 온찜질	30~60분에 걸쳐 천천히 점적 정맥주사	CBC	1. Hypotension: -MILD TO MODERATE: IV fluids -Severe: IV fluids, vasopressor 2. Hypersensitivity reaction: - MILD/MODERATE: Antihistamines with or without inhaled beta agonists, corticosteroids or epinephrine - SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, and IV fluids. 3. Drug-induced dystonia: - ADULT: Benztropine 1 to 2 mg IV or diphenhydramine 1 mg/kg/dose IV over 2 minutes. - CHILD: Diphenhydramine 1 mg/kg/dose IV over 2 minutes	피부노출시 비누와 물로 세척, 점막에 노출 시 물로 씻어냄	투석은 권장하지 않음
39	Fam-trastuzumab deruxtecan-nxki	**ENHERTU**	100mg/V(P)	Anti-HER2 +Topoisomerase I inhibitor		1. IV push 불가 2. 투여시 0.2 - 0.22 μm 크기의 PES 또는 PS 인라인 필터 사용 3. 흔들지 말것, 단회 사용	CBC, left ventricular ejection fraction, pregnancy status, monitor for signs/symptoms of interstitial lung disease/pneumonitis, infusion reactions, cardiovascular monitoring, HBV screening			
40	Fludarabine	**FLUDARA**	50mg/V(P)	Antimetabolite, S phase	Non-vesicants	IVF	CBC, signs and symptoms of tumor lysis syndrome, signs and symptoms of infection	Hemolytic anemia: prednisone 2 mg/kg/day(단, steroids가 효과적인지는 명확하지 않음)	피부나 점막에 접촉되었을 경우에는 물과 비누로 완전히 씻어냄	
41	Fluorouracil	**5-FU**	250mg/5ml/V 500mg/10ml/V 1000mg/20ml/V	Antimetabolite, SG1G2	Irritant, 냉찜질	IV, IVF, IA, 주사시 차광	WBC with differentia, serum glucose in diabetic patients, electrolytes, renal function, and hepatic enzymes	1. Diarrhea: - Uncomplicated mild to moderate diarrhea (Grade 1~2) without signs of infection: loperamide 2 mg every 4 hours for 24 hours.	피부노출시 비누와 물로 세척	투석 비효과적
42	Gemcitabine	**GEMCIBINE** **GEMZAR LIQUID** **GEMTAN SOL**	200mg/V(P) 1g/V(P) 200mg/V 1g/V 200mg/5.26ml/V 1g/26.3ml/V	Antimetabolite	Non-vesicants	1. IVF(30min이상), VS 2. 60분 이상 주입은 독성 증가	CBC with differential and platelet count; renal function tests; hepatic function tests hemolytic uremic syndrome, cardiac monitoring(ECG)		피부나 점막에 접촉되었을 경우에는 물과 비누로 완전히 씻어냄	초기 혈액투석 효과적
43	Gemtuzumab ozogamicin	**MYLOTARG***	4.5mg/V(P)	rhMab(anti CD33)	irritants, 냉찜질	1. 조제부터 투여까지 12hr내 완료 2. 주사시 차광(line은 불필요) 3. 0.2 micron polyethersulfone(PES) filter 사용	ALT, AST, total bilirubin, and alkaline phosphatase, Blood counts, Electrolytes, Veno-occlusive disease, Vital signs and infusion-related reactions, Bleeding, ECG, QT-interval prolongation	권장되는 치료: 과량투여의 경우 일반적인 지지요법을 실시. 혈압과 혈구수를 주의 깊게 관찰		투석되지 않음
44	Idarubicin HCl	**ZAVEL**	5mg/V(P)	Antibiotic	Vesicant, 냉찜질	IV, IVF 1. 주사시 차광(line은 불필요) 2. Heparin과 배합금기	cardiac function, hepatic and renal function	Extravasation 또는 과량투여시 dexrazoxane 투여	가루나 증기를 흡입하거나 피부 또는 눈과 같은 점막에 닿았을 때 눈의 경우 즉시 15분 이상 충분한 물로 씻어낸 후 안과진찰, 피부 접촉시 15분 이상 충분한 물로 씻어낸 후 진찰	Dialysis, hemoperfusion 또는 plasmapheresis은 효과적이지 않음
45	Ifosfamide	**HOLOXAN**	1g/V(P)	Alkylating, CCNS	Irritant, 냉찜질	적어도 30분 이상 천천히 IV infusion	WBC with differential, platelet count, and Hb, glomerular and tubular kidney function, urinalysis, serum and urine chemistries, including phosphorous and potassium, pulmonary toxicity signs and symptoms	1. Bladder toxicity: hydration, diuresis, MESNA 2. Syndrome of inappropriate vasopressin secretion: 1 L/day 이하로 fluid 제한. Furosemide 고려 3. Toxic encephalopathy: methylene blue	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	초기 혈액투석 효과적, hemoperfusion
46	Inotuzumab ozogamicin	**BESPONSA**	1mg/V(P)	CD22-directed ADC	Non-vesicants	IVF(1hr) 주사시 차광(line 차광)	Hepatic venoocclusive disease, including total bilirubin and liver function tests (LFTs), signs and symptoms of infection, bleeding or hemorrhage, QTc interval, CBC, Infusion related reactions (including symptoms such as fever, chills, rash, or breathing problems), ECG	과량 투여 시 주입을 일시적으로 중단, 환자의 간 및 혈액학적 독성에 대해 모니터링		
47	Ipilimumab	**YERVOY**	50mg/10ml/V	Human cytotoxic T-lymphocyte 4 (CTLA-4)-blocking antibody		IVF(90min) 1. 투여시 IV 라인 0.9% NS or 5% DW 로 flush 2. Sterile, non-pyrogenic, low-protein-binding in-line filter 사용	clinical chemistries , liver function tests, thyroid function tests, signs of immune-mediated enterocolitis, bowel perforation such as ileus, of immune-mediated endocrinopathies			
48	Irinotecan Irinotecan liposome	**CALMTOP** **CAMPTO** **ONIVYDE**	40mg/2ml/V 100mg/5ml/V 40mg/2ml/V 100mg/5ml/V 43.3mg/10ml/V	Topoisomerase I inhibitor, S phase	Irritant, 냉찜질	IVF(90min이상) 1. 주사시 차광 2. Cholinergic symptoms 또는 early onset diarrhea 예방, 치료 위해 atropine 0.25~1 mg IV or SubQ 고려 1. 비리포좀화 이리노테칸과 상호 대체해서 사용할 수 없다. 2. 인라인 필터사용 금지	WBC with differential, hemoglobin, and platelet count, diarrhea(early and late onset)	1. Early diarrhea: 24시간 이내 발생. Atropine 투여 2. Late diarrhea: loperamide 3. Fluid and electrolyte therapy 고려	즉시 접촉 부위를 비누와 물로 세척하고 점막 부위를 대량의 물로 씻어냄	
		ERWINASE				1. IM투여 2. Syringe에서 실온 4hr 안정 3. E. coli derived asparaginase hypersensitivity 환자 투여가능	Vital signs (including body temperature), fluid and electrolyte balance, blood glucose, CBC with differential	1. Anaphylaxis: - Mild: antihistamines - Severe: airway management, epinephrine, ECG monitoring, IV	1. 눈에 접촉시 즉시 대량의 water, normal saline, or balanced salt ophthalmologic	

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		
49	L-asparaginase	**LEUNASE**	1만IU/V(P)	Enzyme, G1 phase		IM, IVF, ID(skin test) 1. Skin test : 2U/0.1ml (30 분 이상 관찰) 2. IM: 2ml 초과시 여러부위에 나누어 투여 3. IV: 1hr 이상 IV infousion	and platelet count, serum ammonia concentration, lipase, liver enzymes and renal function.	1. Severe: airway management, epinephrine, ECG monitoring, IV fluids. 2. Body temperature 상승: measures with water and fans, control agitation with benzodiazepines.	saline, or balanced salt ophthalmologic solution으로 15분간 irrigation. 2. 피부 접촉시 즉시 비누와 물로 세척.	
50	Melphalan	**MELPSPAL**	50mg/V(P)	Alkylating, CCNS	Irritant, 냉찜질	1. 중심정맥 투여 2. D5W희석불가, 90분안에 조제투약 완료	CBC	1. Seizures: IV benzodiazepines, 지속 또는 재발 시 barbiturates or propofol 2. Hypersensitivity reaction: - MILD/MODERATE: Antihistamines(inhaled beta agonists, corticosteroids, epinephrine 추가가능) - SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, IV fluids		즉시 Plasmapheresis 고려
51	Methotrexate	**MTX**	50mg/2ml/V 500mg/20ml/V	Antimetabolite (S,G1,G2 phase)	Non-vesicants	1. IV, IVF, IM, IT, SC 2. IT 투여 시 max dose=15mg	CBC with differential and platelets, hepatic function, renal function, renal function by assessing urinary	1. Management of toxicity: 가능한 즉시 IV leucovorin. IV fluids. bicarbonate infusion으로 뇨알칼리화	Antidote: - Leucovorin : 가능한 빨리 노출 1시간 안에	1. High flux hemodialysis , charcoal hemoperfusion 혈액투석이 효과적인지는 알려져있지 않음
52	Mitomycin C	**MITOMYCIN-C**	10mg/V(P)	Antibiotic	Vesicant, 냉찜질	D5W 희석시 사용하면 안정성 : 실온 1~2hr, pH 가 낮은 용액에서는 역가 저하	CBC with differential, renal function tests, liver enzymes, serum electrolytes	Localize by applying cold compresses for 20 minutes 4 times daily for 1 to 2 days. Avoid alcohol compresses. Elevate the limb and	- 피부에 접촉되었을 경우에는 즉시 8.4% 탄산수소나트륨용액으로 여러 번 씻고 비누와 물로	
53	Mitoxantrone HCl	**MITRON**	20mg/10ml/V	TopoisomeraseⅡ inhibitor (S phase)	Irritant, 냉찜질	1. IVF(3min이상) 2. intra-arterial, IM, subQ, IV bolus 불가 3. 뇨변색(청록색), Heparin 배합금기	CBC (including platelets), hepatic function tests, serum uric acid levels, signs and symptoms of tumor lysis syndrome, cardiac function	1. Seizures: IV benzodiazepines, 지속 또는 재발 시 barbiturates or propofol 2. Hypersensitivity reaction: - MILD/MODERATE: Antihistamines with or without inhaled beta agonists, corticosteroids or epinephrine. - SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, and IV fluids	피부에 접촉되었을 때에는 따뜻한 물로 충분히 씻고, 눈에 접촉되었을 때에는 표준 세척방법을 따름	
54	Nelarabine	**ATRIANCE***	250mg/50ml/V	Antimetabolite	Non-vesicants	Hyperuricemia of tumor lysis syndrome(TLS) 예방조치 실시 (eg, hydration, urine alkalinization, and prophylaxis with allopurinol)	CBC(regularly), signs and symptoms of tumor lysis syndrome	1. Seizure: - First line: IV benzodiazepines and/or barbiturates - Refractory seizures: Continuous infusion of propofol, midazolam or pentobarbital 2. Hypotensive episode: IV NS 10 to 20 mL/kg. 적절한 수분 보급 후 저혈압 지속시 dopamine or norepinephrine 사용. Sepsis 여부 평가 시	노출된 피부를 세척하고 눈은 irrigation	
55	Nivolumab	**OPDIVO**	20mg/2ml/V 100mg/10ml/V	Anti-PD1 monoclonal		1. IVF(30min) 2. 투여시 0.2-1.2 μm 크기의 무균,비발열성,저단백결합 인라인	Liver function tests, Serum creatinine levels, Serum creatinine levels,Thyroid function, Signs and symptoms of			
56	Obinutuzumab	**GAZYVA**	1000mg/40ml/V	rhMab(anti CD20)		CLL 환자 첫번째 투여시 100mg 먼저 투여, 문제 없을 경우 900mg 연속으로 투여하고 IRR 발생시 반드시 다음날 투여	Hepatitis B infection, •Liver function, Tumor lysis syndrome, Blood counts, Infusion reactions, Signs or symptoms of developing infection/thrombocytopenia or hemorrhagic events	혈구수에 대한 정기적인 모니터링의 필요, B-세포가 고갈된 환자에서의 감염 위험성 증가에 대해 고려		
57	Oxaliplatin	**ELOXATIN** **OXALITIN** **OXAPLA**	50mg/10ml/V 100mg/20ml/V 50mg/10ml/V 100mg/20ml/V 150mg/30ml/V 50mg/10ml/V 100mg/20ml/V	Alkylating	irritants, 냉or온찜질	1. AI과 반응하여 침전(AI함유의료기기 사용 금지) 2. 주입동안 cold temperatures 피함 3. 다른 약 투여 전 infusion line을 D5W로 flush 4. NaCl을 포함한 모든 농도의 Chloride 물질 및 염기성약물 (5-FU,folinic acid)이나 용액과 배합금기	CBC, liver function tests, serum creatinine, neurologic examination	냉찜질: Apply cold pack for 15 to 20 minutes at least 4 times a day for first 24 hours *treat using a warm compress to avoid the risk of paraesthesia which can be precipitated by cold 온찜질: Localize by apply drug warm compresses for 15-20 minutes 4 times daily for 1 to 2 days. Elevate the limb and administer	피부 또는 점막에 묻은 경우, 즉시 물로 철저하게 씻어냄	Plasma pheresis or plasma exchange 고려. 혈액투석은 권장하지 않음
58	Paclitaxel	**TAXOL** **PADEXOL** **GENEXOL**	30mg/5ml/V 30mg/5ml/V 100mg/16.66ml/V 200mg/33.32ml/V 300mg/50ml/V 30mg/5ml/V 100mg/16.7ml/V	Antimicrotubule(M phase),Plant	Irritant (usually) Vesicant (rare), 냉or온찜질	1. 즉시 접촉 부위를 비누와 물로 세척하고 점막부위를 대량의 물로 씻어냄 2. 0.22 microns 이하의 in-line filter 통해 주입 3. DEHP-함유(PVC) 셋트 사용금지 4. in-line filter set 사용	CBC, vital signs(especially during first hour of infusion), cardiac function, infusion site, liver enzymes, bilirubin, renal function, fluid status and electrolytes	Hyaluronidase: Inject 150 to 900 international units subQ around the extravasation area 냉찜질: Apply cold pack for 15 to 20 minutes at least 4 times a day for first 24 hours 온찜질: Disperse by applying warm compresses for 20 minutes 4 times daily for 1 to 2 days. Elevate the limb and administer	즉시 접촉 부위를 비누와 물로 세척하고 점막 부위를 대량의 물로 씻어냄	과량투여 초기라면, plasmapheresis or plasma exchange은 이론상 효과적일 수 있음
59	Paclitaxel (Albumin 안정제)	**ABRAXANE**	100mg/V(P)	Antimicrotubule (M phase), Plant	Irritant, 냉찜질	in-line filter set 사용 불필요	CBC, Hepatic function, Signs and symptoms of pneumonitis, Infusion site infiltration and possible extravasation	1. Hypersensitivity reaction: MILD/MODERATE: Antihistamines with or without inhaled beta agonists, corticosteroids or epinephrine. SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, and IV fluids. 2. Airway management: Consider orotracheal intubation in patients with significant CNS depression or respiratory failure	Activated charcoal and/or gastric lavage are generally not indicated, ingestion is unlikely	
60	PEG-asparaginase	**ONCASPAR**	3750IU/5ml/V	Enzyme, G1 phase		1. L-aspa hypersensitivity 환자에 투여가능 2. IM: 2ml 초과시 여러부위에 나누어 투여 3. IVF: 1~2hr에 걸쳐 투여(IVF시 100ml혼합)	Coagulation factors, hepatotoxicity, serum glucose, serious allergic reaction or anaphylaxis(for 1 hour after administration), pancreatitis in patients with severe abdominal pain	1. Hypotensive episode: IV 0.9% NaCl, dopamine, or norepinephrine. 2. Allergic rection : - MILD/MODERATE: Antihistamines; inhaled beta agonists for bronchospasm; corticosteroids; epinephrine for severe bronchospasm or angioedema. - SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, ECG monitoring, and IV fluids. 3. Seizure: IV benzodiazepines, barbiturates.		
61	Pembrolizumab	**KEYTRUDA**	100mg/4ml/V	PD-1 blocking antibody		IVF(30min) 투여시 0.2-5 μm 크기의 멸균,비발열성,저단백결합 필터 사용	Liver function, Thyroid function, Hyperglycemia, Renal function			
62	Pemetrexed	**ALIMTA** **PEMED S**	100mg/V(P) 500mg/V(P) 1000mg/40ml/V	Antimetabolite(anti folate), S phase	Non-vesicants	IVF(10min이상) 1. Vit B12, folic acid premedication 필요 2. 피부발진 예방: 경구 dexamethasone	CBC, renal function, hepatic function, plasma homocysteine levels	1. Severe hematologic toxicity : Leucovorin 100 mg/m^2 IV, followed by 50 mg/m^2 IV every 6 hours for 8 days	피부에 닿을 경우 즉시 비누와 물로 피부를 완전히 씻어냄	혈액투석 권장하지 않음
63	Pertuzumab	**PERJETA**	420mg/14ml/V	rhMab (anti-HER2)		1. D5W로 희석 금지 2. 840mg은 60분이상, 420mg은 30~60분 이상 IV infusion(IV push or IV bolus 금기) 3. 연속 투여 시 본제 먼저 투여 후 Docetaxel 투여	Vital signs and fluid status, left ventricular ejection fraction, ECG, infusion reactions; 60 minutes after first infusion and 30 minutes after subsequent infusions	Anaphylaxis: -MILD: Antihistamines -SEVERE: Airway management, epinephrine, cardiac monitoring, IV fluids	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Hemodialysis 권장되지 않음

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		
64	Polatuzumab vedotin	**POLIVY***	140mg/V(P)	Anti-CD79B-directed ADC		1. 투여시 무균, 비발열성, 저단백질 결합 인라인 또는 애드온 필터 (0.2또는 0.22 μm) 및 카테터가 장착된 전용 주입 라인 사용 2. 재구성 및 희석 시 흔들지 말고 부드럽게 조제	CBC, LFTs, pregnancy status, monitor during infusion and for at least 90 minutes after the first infusion, monitor for signs of neuropathy, tumor lysis syndrome, signs/symptoms of progressive multifocal leukoencephalopathy and infection, HBV screening			
65	Pralatrexate	**FOLOTYN***	20mg/1ml/V	Antimetabolite (antifolate)		Vit B_{12}, folic acid premedication 필요	CBC, renal function tests, serum chemistry tests, signs and symptoms of tumor lysis syndrome	MANAGEMENT OF SEVERE TOXICITY: 가능한 빨리 IV leucovorin	피부에 닿을 경우 즉시 비누와 물로 피부를 완전히 씻어냄	1. High flux hemodialysis and charcoal hemoperfusion은 MTX 제거에 도움(특히 신기능 장애 환자) 2. 과량투여 초기에는 혈액투석 강장 추천적
66	Ramucirumab	**CYRAMZA**	100mg/10ml/V 500mg/50ml/V	VEGF-2 receptor targeted antibody		IVF(60min이상) 1. 투여시 0.22㎛ filter사용권장	Proteinuria, Thyroid function•Blood pressure; at least every 2 weeks, Signs and symptoms of infusion-related			
67	Rituximab	**MABTHERA** **TRUXIMA**	100mg/10ml/V 500mg/50ml/V	rhMab (anti CD20)		1. IV push, bolus 금지 2. First infusion	CBC, renal function, signs and symptoms of tumor lysis syndrome, fluid balance, infusion reaction, especially with first infusion and in patients with cardiac or pulmonary conditions	Hypersensitivity reaction: Monitor airway. - MILD/MODERATE: antihistamines with or without inhaled beta agonists, corticosteroids or epinephrine. - SEVERE: Oxygen, aggressive airway management, antihistamines, epinephrine, corticosteroids, ECG monitoring, and IV fluids	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Tumor lysis syndrome 관련 acute renal failure 발생환자는 투석 고려
		MABTHERA SC	1,400mg/11.7ml/V			1. 주사기로 옮긴 후에는 실온 8hr, 냉장 48hr 안정 2. 체표면적과 관계없이 1400mg을 5분간 피하주사. 첫주기에 정맥제제 투여 완료 후 이후 주기부터 피하주사 투여해야 함.				
68	Sacituzumab govitecan-hziy	**TRODELVY(비재고)**	200mg/V(P)	Anti-Trop2 +Topoisomerase I inhibitor		1. 조제전 실온방치, 흔들지 않고 전전히 재구성 및 희석 2. 수액부피 500mL 초과하지 않음 3. IV push 불가, 주사시 차광, 주입완료 후 20ml NS로 주입라인 flush	Blood counts, pregnancy status, monitor for neutropenic fever, hypersensitivity reactions, and infusion-related reactions, HBV screening			
69	Siltuximab	**SYLVANT**	100mg/V	rhMab(anti-IL6 receptor)		IVF(1hr) 1. 희석 후 6시간 이내 주입완료 2. 0.2 micron PES(폴리에테르설폰) 필터가 포함된, PVC, PU(폴리우레탄) 또는 PE 투여세트를 사용	Absolute neutrophil/platelet count, Infection			
70	Streptozocin	**ZANOSAR***	1g/V(P)	Alkylating	Irritant, 냉찜질	Acute insulin release → bed side D50W syringe 비치	CBC, hepatic function, renal function		피부에 닿을 경우 즉시 비누와 물로 피부를 완전히 씻어냄	
71	Tertomotide HCl [Tertomotide]	**RIAVAX***	0.97 mg/V(P) [0.84mg/V(P)]	Cancer immune therapy		Riavax 투여 10~15분 전 Leukine(GM-CSF) 75mcg피내 투여				
72	Thiotepa	**TEPADINA**	15mg/V(P) 100mg/V(P)	Alkylating	Non-vesicants	IV, IVF, VS, IP, IT 투여시 0.22㎛ filter 사용	CBC, platelet count, and fluid and electrolyte status, vital signs		피부에 닿을 경우 즉시 비누와 물로 피부를 완전히 씻어냄	혈액투석
73	Topotecan	**HYCAMTIN**	4mg/V(P)	Topoisomerase I inhibitor (S phase)	Irritant, 냉찜질	IVF(30min이상), IT	CBC, signs and symptoms of interstitial lung disease, fluid and electrolytes and respiratory function		즉시 접촉 부위를 비누와 물로 세척하고 점막 부위를 대량의 물로 씻어냄	
74	Trabectedin	**YONDELIS(비재고)**	1mg//V(P)	Alkylating	Vesicants, 냉찜질	IVF(24hr) 1. Central-line투여 권장(500ml 이상) 2. 말초투여는 적어도 1000ml 이상 희석 3. Dexamethasone premedication필요	CBC, total bilirubin, ALT, AST, alkaline phosphatase, renal function, CPK, pregnancy status, LVEF, Monitor for signs/symptoms of capillary leak syndrome and hypersensitivity reactions, monitor infusion site for signs/symptoms of extravasation, HBV screening		피부, 눈 또는 점막과 접촉했을 경우에는 즉시 다량의 물로 처치	
75	Trastuzumab	**HERCEPTIN** **HERZUMA**	150mg/V(P) 440mg/V(P)	rhMab (anti-HER2)		1. IV infusion only(IV push or IV bolus 금기) 2. 초기투여는 90분 이상 주입, 이후에는 이전 주입에서 이상 없으면 30분 이상 주입 가능	CBC and serum chemistry, signs and symptoms of cardiac dysfunction, hypersensitivity reactions, potential infusion reactions, pulmonary toxicity	1. Anaphylaxis: - MILD: Antihistamines - SEVERE: Airway management, epinephrine, cardiac monitoring and IV fluids. 2. Airway management: Intubate if patient is unable to protect	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	혈액투석 비효과적
		HERCEPTIN SC	600mg/5ml/V			체중과 관계없이 600mg을 2-5분간 SC 주사기로 옮긴 후에는 냉장 48hr 안정, 실온 6hr안정				
76	Trastuzumab emtansine	**KADCYLA**	100mg/V 160mg/V	rhMab (anti-HER2) + antimicrotubule	Non-vesicants	1. 0.22μm polyethersulfone(PES) 필터사용 2. 첫 주입은 90분 이상, 이후에는 30분 이상	Hepatic function, platelet counts, infusion-related reactions, left ventricular ejection fraction	Infusion related reactions(flushing, chills, pyrexia, dyspnea, hypotension, wheezing, bronchospasm, tachycardia)발생 monitoring		
77	Treosulfan	**TRECONDI***	1g/V(P) 5g/V(P)	Alkylating	Vesicants, 냉찜질	filter needle사용 (polytetrafluoroethylen filter)	CBC with differential	Hemorrhagic cystitis: 주입 후 24시간 동안 fluid intake		
		TREOSULFAN*	5g/V(P)							
78	Vinblastine	**VELBASTINE**	10mg/10ml/V	Antimicrotubule (M phase), Plant	Vesicant (Non-DNA-binding), 온찜질	IV 보존제 benzylalcohol 함유	signs and symptoms of extravasation, CBC(mainly WBC), ECG	1. Seizure: IV benzodiazepnes. 지속시 propofol 또는 barbiturate 추가 2. Paralytic ileus: Bowel stimulants, Enema 3. Syndrome of inappropriate vasopressin secretion: fluid restriction, 3% NaCl solution 고려	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	plasmapheresis 또는 exchange transfusion 고려. 혈액투석은 권장하지 않음
79	Vincristine	**VINCRISTINE**	1mg/1ml/V 2mg/2ml/V	M phase, Plant	Vesicant (Non-DNA-binding),	IV, IVF 1. L-asparaginase와 병용 시 12-24hr 전에 투약	serum uric acid levels, blood pressure, CBC	1. Seizure: IV benzodiazepnes. 지속시 propofol 또는 barbiturate 2. Paralytic ileus: Bowel stimulants, Enema		Plasmapheresis 또는 exchange transfusion 고려.
80	Vinorelbine	**NAVELBINE**	10mg/1ml/V 50mg/5ml/V	Antimicrotubule (M phase)	Vesicant (Non-DNA-binding),	1. IV use only: IV(6-10min) up to 30min 2. Intrathecal (IT) 금기	CBC with differential, neuro status. Serum electrolytes	Syndrome of inappropriate vasopressin secretion: fluid 제한. Fluid, electrolytes close monitoring	피부노출시 비누와 물로 세척, 눈에 노출 시 물로 씻어냄	Hemodialysis 권장되지 않음

* : 원내 비입고 항암제, P : powder

Reference: Drug information handbook for oncology, Micromedex, UpToDate, NHS

Elimination
혈액투석 고려

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		

Early hemodialysis 효과적

Hemodialysis or hemoperfusion 고려

Hemodialysis, hemoperfusion and plasmapheresis 권장되지 않음

응급 혈액투석 실시

1-2시간 이내 sodium thiosulfate 투여. large

투석가능 (20% to 50%)

Early hemodialysis 효과적

Early Plasma exchange 고려. 투석 권장하지 않음

투석 권장하지 않음

early hemoperfusion이 배설증가시킴. 투석 권장하지 않음

Early hemoperfusion 고려. Dialysis 권장되지 않음

투석, hemoperfusion or plasmapheresis 권장되지

Dialysis, hemoperfusion, plasmapheresis 권장되지 않음

투석은 권장하지 않음

투석 비효과적

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		

초기 혈액투석 효과적

투석되지 않음

Dialysis, hemoperfusion 또는 plasmapheresis은 효과적이지 않음

조기 혈액투석 효과적, hemoperfusion

즉시 Plasmapheresis 고려

1. High flux hemodialysis , charcoal hemoperfusion 효

혈액투석이 효과적인지는 알려져있지 않음

Plasma pheresis or plasma exchange 고려. 혈액투석은 권장하지 않음

과량투여 초기라면, plasmapheresis or plasma exchange은 이론상 효과적일 수 있음

혈액투석 권장하지 않음

Hemodialysis 권장되지 않

1. High flux hemodialysis and charcoal hemoperfusion은 MTX 제거에 도움(특히 신기능 장애 환자)
2. 과량투여 초기에는 혈액투석 가장 효과적

Tumor lysis syndrome 관련 acute renal failure 발생환자는 투석 고려

Hemodialysis 권장되지 않

1	Alemtuzumab	**LEMTRADA(비재고)**	12mg/1.2ml/V	rhMab(anti CD52)		흔들지 말것, 단회 사용	CBC, serum creatinine, serum transaminases and total bilirubin, urinalysis with urine cell counts, urine protein to creatinine ratio, ECG, TSH, aPTT, signs/symptoms of infection, annual human papillomavirus screening, latent infection screening, signs/symptoms of progressive multifocal leukoencephalopathy and hemophagocytic lymphohistiocytosis, signs/symptoms of thrombotic thrombocytopenic purpura, signs and symptoms of autoimmune encephalitis, baseline and annual skin	과량투여시, 이 약의 투여를 중단하고 부가적인 치료를 중단		혈액투석 - 혈액투석 비효과적 plasmapheresis 또는 exchange transfusion 고려. 혈액투석은 권장하지 않음 Plasmapheresis 또는 exchange transfusion 고려. 혈액투석 권장하지 않음 Hemodialysis 권장되지 않음